

PRABAL BARAHI

Electrical Engineer

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Career Objective

Motivated Electrical Engineering graduate with interests in electrical systems, programming, and software development. Combines a strong foundation in power systems and computational modeling with growing programming skills in Python. Eager to bring engineering discipline and fast adaptability to a dynamic development team as a software intern.

Education

Bachelor in Electrical Engineering

Institute of Engineering (IOE), Pulchowk Campus, Tribhuvan University

Academic Projects

Power Quality Improvement in Isolated AC Microgrids using Spiking Neural Networks for Selective Harmonic Compensation

- Designed and simulated a Shunt Active Power Filter (SAPF) in MATLAB/Simulink to dynamically mitigate grid harmonics.
- Built an event-driven Spiking Neural Network (SNN) controller in Python to replace classical numerical solvers for real-time harmonic compensation.

Illumination and Power Distribution Design of a commercial mall

- Designed lighting layout and low-voltage electrical distribution system of Admin Plaza using AutoCAD.
- Performed load estimation, lux level calculations, detailed electrical installation schematic documentation with single-line diagrams (SLD).

Skills

- Electrical Power Distribution and Installation Design, Power System Analysis and Design, High Voltage Engineering
- MATLAB/Simulink (Intermediate), AutoCAD (Intermediate), Python (Intermediate), Proteus (Beginner), MS Office Tools, C/C++
- Problem Solving, Attention to Detail, Technical Documentation, Adaptability, Time Management, Quick Learner, Team Collaboration

Leadership Experience

Aash Group — Vice Chairperson & Advisor

- Co-founded and helped manage a youth-led NGO; coordinated volunteers, activities, and organizational initiatives during 3 years of service.

Reference

Asst. Prof Anil Kumar Panjiyar

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